Gavin's Lectures on Physics - Quantum Field Theory



Author：长颈鹿 Gavin

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1 Invitation

We work in natural units $\hbar = c = k_B = 1$, with metric $g_{\mu\nu} = \mathrm{diag}(1, -1, -1, -1) = \eta_{\mu\nu}$. This chapter lays out the general classical field–theoretic framework that applies to all fields studied below, then gives a concrete warm–up with the photon polarization vector.

1.1 Polarization Vector

A free electromagnetic plane wave with polarization vector.

$$A^\mu(x) = \epsilon^\mu e^{-ip\cdot x}, \tag{1}$$

where ϵ^μ is the polarization vector.
In Lorenz gauge, the four–divergence vanishes.

$$0 = \partial_\mu A^\mu = -ip_\mu \epsilon^\mu e^{-ip\cdot x},$$

so the Lorenz condition is

$$p_\mu \epsilon^\mu = 0. \tag{2}$$

For a photon moving along the z–axis ($\mathbf{p} = (0, 0, p_z)$),

$$p^\mu = (E_p, 0, 0, p_z), \qquad E_p = |\mathbf{p}| = p_z, \tag{3}$$

so $p^2 = 0$. The Lorenz condition then gives

$$E_p \epsilon^0 - p_z \epsilon^3 = 0.$$

Choosing transverse/Coulomb gauge $\epsilon^0 = \epsilon^3 = 0$,

$$\epsilon^\mu = (0, \epsilon_x, \epsilon_y, 0).$$

The transverse complex amplitudes form the Jones vector.

$$\mathbf{J} = \begin{pmatrix} \epsilon_x \\ \epsilon_y \end{pmatrix}, \tag{4}$$

often normalized as $\mathbf{J}^\dagger \mathbf{J} = 1$. The physical electric field is the real part of the complex amplitude. The formal quantization of the vector field is treated in Chapter 4.

1.2 $e^+e^- \to \mu^+\mu^-$ as a Preview

The virtual photon's right–handed circular polarization vector for a photon propagating along the z–axis is $\epsilon^\mu = (0, 1, i, 0)$.

$$\langle \gamma | H_I | e^+e^- \rangle^\mu \propto e\epsilon^\mu = e(0, 1, i, 0) \tag{5}$$

Rotate angle θ in the xz-plane.

$$\langle \gamma | H_I | e^+e^- \rangle^\mu \propto e(0, \cos\theta, i, -\sin\theta) \tag{6}$$

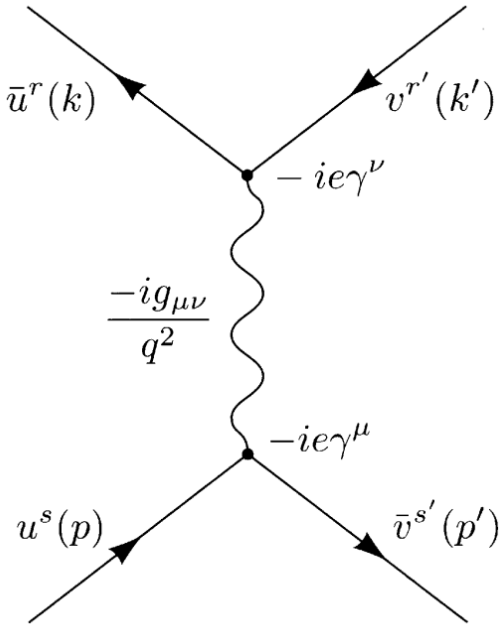


Figure 1: Tree–level Feynman diagram for $e^+e^- \to \mu^+\mu^-$ via a virtual photon.

Applying the Feynman rules to the tree–level diagram below, the invariant amplitude factors into the electron current, the photon propagator, and the muon current. The Feynman rules directly give $i\mathcal{M}$:

$$\begin{aligned} i\mathcal{M} &= \bar{v}^{s'}(p')(-ie\gamma^\mu)u^s(p)\left(\frac{-ig_{\mu\nu}}{q^2}\right)\bar{u}^r(k)(-ie\gamma^\nu)v^{r'}(k') \\ &= \frac{ie^2}{q^2}\left(\bar{v}^{s'}(p')\gamma^\mu u^s(p)\right)\left(\bar{u}^r(k)\gamma_\mu v^{r'}(k')\right), \end{aligned} \tag{7}$$

so the invariant amplitude itself is

$$\mathcal{M} = \frac{e^2}{q^2}\left(\bar{v}^{s'}(p')\gamma^\mu u^s(p)\right)\left(\bar{u}^r(k)\gamma_\mu v^{r'}(k')\right). \tag{8}$$

Here $q = p + p' = k + k'$. Choosing the Weyl spinors for specific helicities (R/L = right/left–handed) and evaluating the bilinears $\bar{v}\gamma^\mu u$ yields the four helicity amplitudes. By way of illustration, for $RL \to RL$:

$$\mathcal{M}(RL \to RL) \sim \langle\mu^+\mu^-|H_I|\gamma\rangle^\mu\langle\gamma|H_I|e^+e^-\rangle_\mu = -e^2(1+\cos\theta) \tag{9}$$

The remaining three follow by the same substitution:

$$\mathcal{M}(LR \to LR) = -e^2(1+\cos\theta) \tag{10}$$

$$\mathcal{M}(LR \to RL) = -e^2(1-\cos\theta) \tag{11}$$

$$\mathcal{M}(RL \to LR) = -e^2(1-\cos\theta) \tag{12}$$

Average for square of the 4 conditions:

$$|\mathcal{M}|^2 = e^4(1 + \cos^2\theta) \tag{13}$$

Fine-structure constant:

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137} \tag{14}$$

E_{cm} is energy of the system in center of mass frame.
Differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 E_{cm}^2} \cdot |\mathcal{M}|^2 = \frac{\alpha^2}{4E_{cm}^2}(1 + \cos^2\theta) \tag{15}$$

Total cross section,

$$\sigma = \frac{4\pi\alpha^2}{3E_{cm}^2} \tag{16}$$

1.3 Action Principle

The action is the spacetime integral of the Lagrangian density.

$$S = \int \mathrm{d}^4x\, \mathcal{L}(\phi, \partial_\mu\phi) \tag{17}$$

Under a coordinate and field variation,

$$\delta\mathcal{L} = \mathcal{L}'(x') - \mathcal{L}(x), \qquad \delta(\,\mathrm{d}^4x) = \,\mathrm{d}^4x\,(\partial_\mu \delta x^\mu) \tag{18}$$

Varying the action and dropping boundary terms,

$$\begin{aligned} 0 = \delta S = \int \mathrm{d}^4x \Bigg\{ &\left[\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right] \bar{\delta}\phi \\ &+\partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \cdot \bar{\delta}\phi + \mathcal{L}\,\delta x^\mu\right]\Bigg\} \end{aligned} \tag{19}$$

The Lagrangian and the field variation split as follows.

$$L(t) = \int \mathrm{d}^3x\, \mathcal{L}(\phi, \partial_\mu\phi), \qquad \delta\phi = \bar{\delta}\phi + \partial_\mu\phi\, \delta x^\mu \tag{20}$$

1.3.1 Euler–Lagrange Equation

Stationarity of the action gives the field equation.

$$\partial_\mu \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} - \frac{\partial\mathcal{L}}{\partial\phi} = 0 \tag{21}$$

1.3.2 Hamiltonian

Canonical momentum and Hamiltonian density.

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}, \qquad \mathcal{H} = \pi\dot{\phi} - \mathcal{L}, \qquad H = \int \mathrm{d}^3x\, \mathcal{H} \tag{22}$$

Canonical equations of motion.

$$\dot{\pi} = -\frac{\partial \mathcal{H}}{\partial \phi}, \qquad \dot{\phi} = \frac{\partial \mathcal{H}}{\partial \pi} \tag{23}$$

1.3.3 Noether Theorem

Every continuous symmetry yields a conserved current. (Convention: $\bar{\delta}\phi$ is the change at a fixed spacetime point, δx^μ the coordinate shift; overall sign depends on this choice.)

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\, \bar{\delta}\phi + \mathcal{L}\, \delta x^\mu, \qquad \partial_\mu j^\mu = 0 \tag{24}$$

Divergence theorem converts local conservation into a boundary statement.

$$0 = \int \mathrm{d}^3x\, \partial_\mu j^\mu = \int \mathrm{d}^3x\, \frac{\partial j^0}{\partial t} + \int \mathrm{d}^3x\, \nabla \cdot \mathbf{j} \tag{25}$$

The conserved charge, time independent by Gauss's law.

$$Q = \int \mathrm{d}^3x\, j^0, \qquad \frac{dQ}{dt} = -\int \mathrm{d}^3x\, (\nabla \cdot \mathbf{j}) = -\int^{\infty} \mathbf{j} \cdot\, \mathrm{d}\sigma = 0 \tag{26}$$

The formalism above applies to any field; we apply it first to the scalar field in Chapter 2, then reuse it for spinor and vector fields.

2 Scalar Field

We begin with the simplest field: the real scalar field, which describes spin–0 particles. Throughout, $E_p = \sqrt{\mathbf{p}^2 + m^2}$ and $p^\mu = (E_p, \mathbf{p})$.

2.1 Klein–Gordon Equation

The free scalar Lagrangian.

$$\mathcal{L} = \frac{1}{2}\left(\partial_\mu\phi\,\partial^\mu\phi - m^2\phi^2\right) \tag{27}$$

Canonical momentum and Hamiltonian density.

$$\pi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi}, \qquad \mathcal{H} = \frac{1}{2}\left[\pi^2 + (\nabla\phi)^2 + m^2\phi^2\right] \tag{28}$$

The Klein–Gordon equation and the d'Alembertian.

$$(\partial^2 + m^2)\phi = 0, \qquad \partial^2 = \partial_\mu\partial^\mu = \partial_0^2 - \nabla^2 \tag{29}$$

2.1.1 Plane Wave Solutions

Mode expansion in the standard relativistic normalization.

$$\phi(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\left(a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x}\right) \tag{30}$$

Klein–Gordon inner product (orthogonality).

$$\int \mathrm{d}^3x\, f_p^*(x)\, i\overleftrightarrow{\partial}_0\, f_{p'}(x) = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{p}'), \qquad f_p(x) = \frac{e^{-ip\cdot x}}{\sqrt{2E_p}}, \quad A\overleftrightarrow{\partial}_\mu B = A\partial_\mu B - (\partial_\mu A)B \tag{31}$$

Relativistic phase–space measure identity.

$$\int \frac{\mathrm{d}^3p}{2E_p} = \int \mathrm{d}^4p\,\delta(p^2 - m^2)\,\theta(p^0) \tag{32}$$

2.2 Canonical Quantization

Equal–time canonical commutation relations.

$$[\phi(\mathbf{x},t),\phi(\mathbf{x}',t)] = 0, \qquad [\pi(\mathbf{x},t),\pi(\mathbf{x}',t)] = 0, \qquad [\phi(\mathbf{x},t),\pi(\mathbf{x}',t)] = i\delta^{(3)}(\mathbf{x}-\mathbf{x}') \tag{33}$$

Creation and annihilation operators satisfy harmonic oscillator algebra.

$$[a_p, a_{p'}] = 0, \qquad [a_p^\dagger, a_{p'}^\dagger] = 0, \qquad [a_p, a_{p'}^\dagger] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{p}') \tag{34}$$

2.2.1 Box Normalization

Replace continuum integrals by discrete sums in a box $V = L^3$:

$$\int \frac{\mathrm{d}^3 p}{(2\pi)^3} \to \sum_{\mathbf{p}} \frac{1}{V}, \qquad \delta^{(3)}(\mathbf{p} - \mathbf{p}') \to \frac{V}{(2\pi)^3} \delta_{\mathbf{p}\mathbf{p}'} \tag{35}$$

Mode expansion in finite volume.

$$\phi(x) = \sum_{\mathbf{p}} \frac{1}{\sqrt{2E_p V}} \left(a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x} \right) \tag{36}$$

2.2.2 Particle Number Representation

In box normalization the commutator becomes discrete: $[a_p, a_{p'}^\dagger] = \delta_{pp'}$, which makes N_p a discrete number operator with integer eigenvalues.

Number operator and its eigenstates.

$$N_p = a_p^\dagger a_p, \qquad N_p \left| n_p \right\rangle = n_p \left| n_p \right\rangle \tag{37}$$

Raising and lowering on the number–state ladder.

$$a_p^\dagger \left| n_p \right\rangle = \sqrt{n_p + 1} \left| n_p + 1 \right\rangle, \qquad a_p \left| n_p \right\rangle = \sqrt{n_p} \left| n_p - 1 \right\rangle \tag{38}$$

Vacuum condition and normalized number states.

$$a_p \left| 0 \right\rangle = 0, \qquad \left| n_p \right\rangle = \frac{(a_p^\dagger)^{n_p}}{\sqrt{n_p!}} \left| 0 \right\rangle \tag{39}$$

2.2.3 Energy–Momentum Tensor

The symmetric Noether current for spacetime translations.

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - \mathcal{L} g^{\mu\nu}, \qquad \partial_\mu T^{\mu\nu} = 0 \tag{40}$$

Total four–momentum as the spatial integral of $T^{0\nu}$.

$$P^\nu = \int \mathrm{d}^3 x \, T^{0\nu}, \qquad T^{00} = \mathcal{H} = \frac{1}{2} \left[\pi^2 + (\nabla\phi)^2 + m^2 \phi^2 \right], \qquad P^0 = H = \int \mathrm{d}^3 x \, \mathcal{H} \tag{41}$$

Mode expansion of the four–momentum (in box normalization; continuum version follows by replacing $\sum_{\mathbf{p}} \to \int \frac{\mathrm{d}^3 p}{(2\pi)^3} V$).

$$P^\mu = \int \mathrm{d}^3 x \, T^{0\mu} = \sum_{\mathbf{p}} p^\mu \left(N_p + \frac{1}{2} \right) \tag{42}$$

2.2.4 Single–Particle Wave Function

One–particle wave function as the field's matrix element.

$$\psi_{\mathbf{p}} = \frac{1}{\sqrt{2E_p}} \langle \mathbf{p}|\psi\rangle = \langle 0|\, a_p\, |\psi\rangle\,, \qquad \psi(\mathbf{x},t) = \langle 0|\, \phi(x,t)\, |\psi\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} e^{-ip\cdot x} \psi_{\mathbf{p}} \tag{43}$$

2.2.5 Zero–Point Energy

Vacuum energy summed over all modes, divergent.

$$E_0 = \frac{V}{(2\pi)^3} \int \mathrm{d}^3p \frac{1}{2} E_p = \frac{V}{16\pi^3} \int_0^{K_c} 4\pi p^2 \sqrt{p^2+m^2}\, \mathrm{d}p \sim K_c^4 \tag{44}$$

This diverges as K_c^4 (ultraviolet divergence). It is removed by normal ordering: $H \to H - E_0 = \sum_{\mathbf{p}} N_p E_p$.

2.2.6 Fock Space

Vacuum and relativistically normalized one–particle states.

$$|\mathbf{p}\rangle = \sqrt{2E_p}\, a_p^\dagger\, |0\rangle\,, \qquad \langle \mathbf{p}|\mathbf{q}\rangle = 2E_p (2\pi)^3 \delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{45}$$

Two–particle states are symmetric (bosons). The number operator is $N = \int \frac{\mathrm{d}^3p}{(2\pi)^3} a_p^\dagger a_p$.

2.3 Complex Scalar Field

Lagrangian for a charged (complex) scalar.

$$\mathcal{L} = \partial_\mu \phi^\dagger\, \partial^\mu \phi - m^2 \phi^\dagger \phi \tag{46}$$

U(1) symmetry gives the conserved current and charge.

$$j^\mu = i(\phi^\dagger \partial^\mu \phi - \phi \partial^\mu \phi^\dagger), \qquad Q = \int \mathrm{d}^3x\, j^0 \tag{47}$$

Mode expansion with particle and antiparticle operators.

$$\phi(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \left(a_p e^{-ip\cdot x} + b_p^\dagger e^{ip\cdot x} \right) \tag{48}$$

Hamiltonian and conserved charge in modes (already normal ordered; zero-point energy subtracted).

$$H = \int \frac{\mathrm{d}^3p}{(2\pi)^3} E_p (a_p^\dagger a_p + b_p^\dagger b_p), \qquad Q = \int \frac{\mathrm{d}^3p}{(2\pi)^3} (a_p^\dagger a_p - b_p^\dagger b_p) = N_a - N_b \tag{49}$$

where $N_a = \int \frac{\mathrm{d}^3p}{(2\pi)^3} a_p^\dagger a_p$ and $N_b = \int \frac{\mathrm{d}^3p}{(2\pi)^3} b_p^\dagger b_p$.

The negative–frequency solutions correspond to antiparticles. For a real scalar field the particle is its own antiparticle; for a complex scalar field they form particle–antiparticle pairs.

2.4 Causality and Feynman Propagator

2.4.1 Causality and Spacelike Separation

Spacelike separation criterion.

$$(x-y)^2 = (t_1-t_2)^2 - |\mathbf{x}_1-\mathbf{x}_2|^2 \tag{50}$$

For spacelike separations $(x-y)^2 < 0$, signals cannot propagate; for timelike or lightlike separations $(x-y)^2 \geq 0$, signals can propagate.

2.4.2 Vacuum Two–Point Function

Wightman function, positive–frequency part only.

$$D(x-y) = \langle 0|\phi(x)\phi(y)|0\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p}e^{-ip\cdot(x-y)} \tag{51}$$

Commutator vanishes outside the light cone by cancellation.

$$[\phi(x),\phi(y)] = D(x-y) - D(y-x) \tag{52}$$

2.4.3 Feynman Propagator

Time–ordered product definition.

$$T\phi(x)\phi(y) = \theta(x^0-y^0)\phi(x)\phi(y) + \theta(y^0-x^0)\phi(y)\phi(x) \tag{53}$$

The Feynman propagator in momentum space.

$$D_F(x-y) = \langle 0|T\phi(x)\phi(y)|0\rangle = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{i}{p^2-m^2+i\epsilon}e^{-ip\cdot(x-y)} \tag{54}$$

Green's function for the Klein–Gordon operator.

$$(\partial^2+m^2)D_F(x-y) = -i\delta^{(4)}(x-y) \tag{55}$$

The $i\epsilon$ prescription shifts the poles off the real axis to make the integral well–defined; one takes $i\epsilon \to 0$ at the end.

3 Spinor Field

We now turn to the Dirac field, which describes spin–$\frac{1}{2}$ particles such as the electron.

3.1 Dirac Equation and Gamma Matrices

Factorize the KG operator into first–order derivatives.

$$0 = (\eta^{\mu\nu}\partial_\mu\partial_\nu + m^2)\psi = [-\eta^{\mu\nu}(i\partial_\mu)(i\partial_\nu) + m^2]\psi \tag{56}$$

Introducing the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$,

$$0 = -(i\gamma^\mu\partial_\mu + m)(i\gamma^\nu\partial_\nu - m)\psi \tag{57}$$

The Dirac equation in slashed notation.

$$\text{Def:} \quad \not{\partial} = \gamma^\mu\partial_\mu, \qquad (i\not{\partial} - m)\psi = 0 \tag{58}$$

3.1.1 Dirac Matrices in Chiral Representation

Pauli matrices (building blocks of γ^i).

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{59}$$

Gamma matrices in the chiral (Weyl) representation.

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \tag{60}$$

Chiral gamma matrix.

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I_2 & 0 \\ 0 & I_2 \end{pmatrix} \tag{61}$$

Chiral projection operators.

$$P_L = \frac{1-\gamma^5}{2} = \begin{pmatrix} I_2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_R = \frac{1+\gamma^5}{2} = \begin{pmatrix} 0 & 0 \\ 0 & I_2 \end{pmatrix} \tag{62}$$

Properties: $\gamma^{0\dagger} = \gamma^0$, $\gamma^{i\dagger} = -\gamma^i$, and $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$.

3.2 Plane Wave Solutions

Adjoint spinor and Dirac Lagrangian.

$$\bar{\psi} = \psi^\dagger\gamma^0, \qquad \mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi \tag{63}$$

Positive–frequency (particle) solutions.

$$\psi(x) = u(p)e^{-ip\cdot x}, \qquad (\not{p} - m)u(p) = 0 \tag{64}$$

Negative–frequency (antiparticle) solutions.

$$\psi(x) = v(p)e^{ip\cdot x}, \qquad (\not{p} + m)v(p) = 0 \tag{65}$$

Spinor normalization conventions (spinor indices contracted).

$$u^\dagger u = v^\dagger v = 2E_p, \qquad \bar{u}u = -\bar{v}v = 2m \tag{66}$$

Completeness relations over spin states.

$$\sum_s u^s(p)\bar{u}^s(p) = \not{p} + m, \qquad \sum_s v^s(p)\bar{v}^s(p) = \not{p} - m \tag{67}$$

Lorentz generator in the spinor representation.

$$S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu] \tag{68}$$

3.3 Quantization and Spin–Statistics

Canonical momentum and mode expansion.

$$\psi(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{s=1,2} \left[a_p^s u^s(p) e^{-ip\cdot x} + b_p^{s\dagger} v^s(p) e^{ip\cdot x}\right] \tag{69}$$

Hamiltonian in modes (already normal ordered; the fermion vacuum energy has been subtracted).

$$H = \int \frac{\mathrm{d}^3p}{(2\pi)^3} E_p \sum_s (a_p^{s\dagger} a_p^s + b_p^{s\dagger} b_p^s) \tag{70}$$

Canonical anticommutation relations.

$$\{a_p^r, a_q^{s\dagger}\} = \{b_p^r, b_q^{s\dagger}\} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q})\delta^{rs} \tag{71}$$

Equal–time anticommutators for the field itself.

$$\{\psi_a(x), \psi_b^\dagger(y)\} = \delta^{(3)}(\mathbf{x} - \mathbf{y})\delta_{ab}, \qquad \{\psi, \psi\} = \{\psi^\dagger, \psi^\dagger\} = 0 \tag{72}$$

Scalar fields satisfy $[a_p^\dagger, a_q^\dagger] = 0$ (symmetric under exchange, bosons); spinor fields satisfy $\{a_p^\dagger, a_q^\dagger\} = 0$ (antisymmetric under exchange, fermions).

3.4 Fermion Propagator

Time–ordered product with fermionic minus sign.

$$T\psi(x)\bar{\psi}(y) = \theta(x^0 - y^0)\psi(x)\bar{\psi}(y) - \theta(y^0 - x^0)\bar{\psi}(y)\psi(x) \tag{73}$$

The Feynman propagator for the Dirac field.

$$S_F(x - y) = \langle 0|T\psi(x)\bar{\psi}(y)|0\rangle = \int \frac{\mathrm{d}^4p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} e^{-ip\cdot(x-y)} \tag{74}$$

Green's function for the Dirac operator.

$$(i\ \not{\partial} - m)S_F(x - y) = i\delta^{(4)}(x - y) \tag{75}$$

4 Vector Field

The vector field describes spin–1 particles. We first treat the massless case (the electromagnetic field), then note the massive (Proca) case for comparison.

4.1 Maxwell Field and Gauge Invariance

Field strength tensor (upper–index convention, Peskin & Schroeder).

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E^1 & 0 & B^3 & -B^2 \\ -E^2 & -B^3 & 0 & B^1 \\ -E^3 & B^2 & -B^1 & 0 \end{pmatrix} \tag{76}$$

Maxwell Lagrangian.

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2}(E^2 - B^2) \tag{77}$$

Gauge redundancy $A_\mu \to A_\mu + \partial_\mu \Lambda$. In Coulomb gauge $\nabla \cdot \mathbf{A} = 0$. Source–free Maxwell equations.

$$\partial_\mu F^{\mu\nu} = 0 \tag{78}$$

4.2 Proca Equation: Massive Vector Field

Lagrangian for a massive vector field.

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 A_\mu A^\mu. \tag{79}$$

The Proca equation of motion.

$$\partial_\mu F^{\mu\nu} + m^2 A^\nu = 0. \tag{80}$$

Taking the divergence yields the Lorenz condition automatically.

$$m^2 \, \partial_\nu A^\nu = 0 \quad \Rightarrow \quad \partial_\mu A^\mu = 0 \quad (m \neq 0). \tag{81}$$

Thus for $m \neq 0$ the Lorenz condition holds automatically and there is no gauge freedom; a massive vector has **three** physical polarization degrees of freedom. For $m = 0$ (photon), gauge invariance remains and only **two** transverse polarizations are physical.

4.2.1 Polarization Vectors

Massive vector polarization basis.

$$p \cdot \epsilon_\lambda = 0, \qquad \epsilon_\lambda \cdot \epsilon_{\lambda'} = -\delta_{\lambda\lambda'}. \tag{82}$$

Completeness relation.

$$\sum_{\lambda=1}^{3} \epsilon_\lambda^\mu(p) \epsilon_\lambda^{\nu *}(p) = -g^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}. \tag{83}$$

For a massless photon, only two transverse polarizations $\lambda = 1, 2$ remain physical; the longitudinal and scalar modes are gauge artifacts.

4.3 Mode Expansion

Photon mode expansion with two physical transverse polarizations (Coulomb gauge). In covariant quantization one includes all four polarizations and removes the unphysical longitudinal/scalar modes via the Gupta–Bleuler / BRST condition; the result is equivalent to the two–polarization expansion below.

$$A^\mu(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{\lambda=1,2} \left[a_p^\lambda \epsilon^{\lambda\mu}(p) e^{-ip\cdot x} + a_p^{\lambda\dagger} \epsilon^{\lambda *\mu}(p) e^{ip\cdot x} \right] \tag{84}$$

Photon commutation relations.

$$[a_p^\lambda, a_{p'}^{\lambda'\dagger}] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta^{\lambda\lambda'} \tag{85}$$

4.4 Covariant Gauge and Propagator

Gauge–fixing term.

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2\xi} (\partial^\mu A_\mu)^2 \tag{86}$$

The R_ξ gauge propagator (where q denotes the propagator momentum, distinct from external particle momentum p).

$$D_F^{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon} \left[g^{\mu\nu} - (1-\xi) \frac{q^\mu q^\nu}{q^2 + i\epsilon} \right] \tag{87}$$

In Feynman gauge $\xi = 1$: $D_F^{\mu\nu} = -ig^{\mu\nu}/(q^2 + i\epsilon)$. The Ward identity $q^\mu M_\mu = 0$ ensures that the unphysical longitudinal/scalar modes cancel in any physical amplitude.

5 Interacting Field

We now turn to interactions. The free–field propagators derived in Chapters 2–4 are the building blocks; interactions introduce vertices and connect them.

5.1 Interaction Lagrangians

5.1.1 ϕ^4 Theory

The ϕ^4 interaction and its equation of motion. (Here λ is the coupling constant, not the polarization index used in Chapter 4.)

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4, \qquad (\partial^2 + m^2)\phi = -\frac{\lambda}{3!}\phi^3 \tag{88}$$

5.1.2 QED

QED interaction and covariant derivative.

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu, \qquad D_\mu = \partial_\mu + ieA_\mu(x) \tag{89}$$

Full QED Lagrangian.

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}(F_{\mu\nu})^2 \tag{90}$$

Equations of motion: $(i\not{D} - m)\psi = 0$, and $\partial_\mu F^{\mu\nu} = ej^\nu$ with $j^\nu = \bar{\psi}\gamma^\nu\psi$.

5.2 Interaction Picture

Three pictures compared (assuming operators have no explicit time dependence).

Picture	State evolution driven by	Operator evolution driven by
Schrödinger	H (full Hamiltonian)	none (O_s time–independent)
Heisenberg	none (state time–independent)	H (full Hamiltonian)
Interaction	H_I (interaction Hamiltonian)	H_0 (free Hamiltonian)

Interaction Hamiltonian.

$$H_I = e^{iH_0 t} H_{\text{int}} e^{-iH_0 t}, \qquad H_{\text{int}} = -\int \mathrm{d}^3x \mathcal{L}_{\text{int}} \tag{91}$$

5.3 Dyson Series and Gell–Mann–Low Formula

Schrödinger equation in the interaction picture.

$$i\frac{\partial \left|\psi(t)\right\rangle_I}{\partial t} = H_I(t)\left|\psi(t)\right\rangle_I, \qquad \left|\psi(t)\right\rangle_I = U_I(t,t_0)\left|\psi(t_0)\right\rangle_I \tag{92}$$

Dyson series (time–ordered exponential).

$$U_I(t,t_0) = \mathcal{T}\left\{\exp\left[-i\int_{t_0}^{t}\mathrm{d}t'\, H_I(t')\right]\right\} \tag{93}$$

$|0\rangle$: free vacuum (ground state of H_0); $|\Omega\rangle$: interacting vacuum (ground state of H).

Gell–Mann–Low formula.

$$\langle\Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|T\{\phi_I(x_1)\cdots\phi_I(x_n)\, e^{-i\int_{-T}^{T}\mathrm{d}tH_I(t)}\}|0\rangle}{\langle 0|T\{e^{-i\int_{-T}^{T}\mathrm{d}tH_I(t)}\}|0\rangle} \tag{94}$$

5.4 The S–Matrix

The S–matrix element is the overlap between asymptotically free initial and final states.

$$S_{fi} = \langle f|S|i\rangle. \tag{95}$$

Taking $t_0 \to -\infty$ and $t \to +\infty$ with adiabatic switching of the interaction, the S–matrix is the limit of the Dyson series of Section 5.3:

$$S = U_I(\infty,-\infty) = \mathcal{T}\exp\left[-i\int_{-\infty}^{\infty}\mathrm{d}t\, H_I(t)\right], \tag{96}$$

where $H_I(t)$ is the interaction Hamiltonian in the interaction picture and $\mathcal{T}$ denotes time ordering.

5.4.1 T–Matrix Decomposition

Separate the identity (no–scattering) part from the genuine transition part.

$$S_{fi} = \delta_{fi} + iT_{fi}. \tag{97}$$

The T–matrix factorizes out overall four–momentum conservation, leaving the Lorentz–invariant amplitude $\mathcal{M}_{fi}$:

$$T_{fi} = (2\pi)^4\delta^{(4)}(p_f - p_i)\,\mathcal{M}_{fi}, \tag{98}$$

so that

$$S_{fi} = \delta_{fi} + i(2\pi)^4\delta^{(4)}(p_f - p_i)\,\mathcal{M}_{fi}. \tag{99}$$

For $i \neq f$, the δ_{fi} term vanishes, leaving the transition amplitude. The invariant amplitude $\mathcal{M}_{fi}$ is obtained from the Feynman rules of Section 5.6.

5.4.2 Transition Probability: Discrete vs. Continuum

For discrete, finite–volume normalized final states, the transition probability is

$$P_{i\to f} = |S_{fi}|^2 \qquad (i \neq f). \tag{100}$$

For continuum momentum states, S_{fi} contains $(2\pi)^4\delta^{(4)}(p_f - p_i)$, so $|S_{fi}|^2$ contains $[\delta^{(4)}]^2$ and is not an ordinary probability. It must be regularized:

$$\left|(2\pi)^4\delta^{(4)}(p_f - p_i)\right|^2 \;\longrightarrow\; VT\,(2\pi)^4\delta^{(4)}(p_f - p_i), \tag{101}$$

where V is the spatial volume and T the time interval. In practice, for continuum states one uses the transition rate $\mathrm{d}\Gamma$, the probability element $\mathrm{d}P$, or the cross section $\mathrm{d}\sigma$ (defined in Section 5.7), integrated over phase space.

5.5 Wick's Theorem

Normal ordering N: move all $a^\dagger$ to the left and all a to the right. Split the field into positive/negative frequency parts.

$$\phi_I(x) = \phi_I^+(x) + \phi_I^-(x), \qquad \phi_I^+(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{a_p}{\sqrt{2E_p}} e^{-ip\cdot x} \tag{102}$$

Contraction equals the time–ordered product's vacuum expectation.

$$D_F(x-y) = \underbrace{\phi(x)\phi(y)} = \langle 0|T\phi(x)\phi(y)|0\rangle \tag{103}$$

Wick's theorem (schematic; the actual expansion sums over all possible complete sets of contractions). For fermions, each exchange of fermionic fields picks up a minus sign.

$$T\{\phi(x_1)\cdots\phi(x_m)\} = N\left\{\phi(x_1)\cdots\phi(x_m) + \sum_{\text{all complete contraction sets}} \underbrace{\phi(x_i)\phi(x_j)}\ \phi(x_k)\cdots\right\} \tag{104}$$

5.6 Feynman Rules

Wick's theorem (Section 5.5) reduces the time–ordered products in the Gell–Mann–Low formula to contractions, i.e. propagators, joined at interaction vertices. The resulting Feynman rules are:

Position–space (ϕ^4): propagator $D_F(x-y)$; vertex $(-i\lambda)\int \mathrm{d}^4 z$; external leg 1; divide by symmetry factor.

Momentum–space (ϕ^4):

- Propagator: $\dfrac{i}{p^2 - m^2 + i\epsilon}$
- Vertex: $-i\lambda$
- External leg: 1 (scalar; replaced by spinors/polarizations for other fields)
- Momentum conservation: $(2\pi)^4\delta^{(4)}(\sum p)$
- Loop integral: $\int \frac{\mathrm{d}^4 p}{(2\pi)^4}$
- Divide by symmetry factor.

5.7 Cross Sections and Decay Rates

With the Feynman rules of Section 5.6, we can compute $\mathcal{M}$ for any process and extract physical observables.

5.7.1 Differential Cross Section

Starting from the transition probability $P_{i\to f} = |S_{fi}|^2$ of Section 5.4.2, dividing by the flux factor and the time interval T, and using the regularization $|(2\pi)^4\delta^{(4)}|^2 \to VT(2\pi)^4\delta^{(4)}$, one obtains the differential cross section for $2 \to n$ scattering:

$$\mathrm{d}\sigma = \frac{1}{4E_1E_2|v_1 - v_2|}|\mathcal{M}|^2\,(2\pi)^4\delta^{(4)}(p_f - p_i)\;\mathrm{d}\Phi_n, \tag{105}$$

where the final–state Lorentz–invariant phase space is

$$\mathrm{d}\Phi_n = \prod_f \frac{\mathrm{d}^3p_f}{(2\pi)^3\,2E_f}. \tag{106}$$

For collinear beams, the flux factor can also be written in Lorentz–invariant form:

$$4E_1E_2|v_1 - v_2| = 4\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2}. \tag{107}$$

If initial spins or polarizations are not measured, average over them:

$$|\mathcal{M}|^2 \;\longrightarrow\; \frac{1}{(2s_1+1)(2s_2+1)}\sum_{\text{spins}}|\mathcal{M}|^2. \tag{108}$$

For identical particles in the final state, include the appropriate symmetry factor (e.g. $1/n!$ for n identical particles).

5.7.2 Total Cross Section and Decay Rate

The total cross section is the integral over final–state phase space.

$$\sigma = \int \mathrm{d}\sigma. \tag{109}$$

The decay rate Γ is the probability per unit time for an unstable particle to decay:

$$\Gamma = \frac{\text{decays per unit time}}{\text{number of } A \text{ particles}}. \tag{110}$$

5.7.3 Center–of–Mass Frame for $2 \to 2$ Scattering

For $2 \to 2$ scattering in the CM frame,

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{cm}} = \frac{1}{64\pi^2 s}\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2, \qquad s = E_{\text{cm}}^2, \tag{111}$$

where $E_{\text{cm}} = \sqrt{s}$ is the **total** center–of–mass energy.

For massless initial and final states (or equal incoming/outgoing masses), $|\mathbf{p}_f|/|\mathbf{p}_i| \to 1$, so

$$\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)_{\text{cm}} = \frac{|\mathcal{M}|^2}{64\pi^2 s} = \frac{|\mathcal{M}|^2}{64\pi^2 E_{\text{cm}}^2}. \tag{112}$$

This is the formula used in the $e^+e^- \to \mu^+\mu^-$ calculation in Chapter 1.

5.7.4 Breit–Wigner Resonance

Near a resonance energy E_R, the cross section follows the Breit–Wigner form.

$$f(E) \propto \frac{1}{E - E_R + i\Gamma/2}, \qquad \sigma \propto |f(E)|^2 \propto \frac{1}{(E - E_R)^2 + \Gamma^2/4} \tag{113}$$

5.8 Path Integral Formulation

The time–evolution operator generates the propagator kernel.

$$|\psi(t)\rangle = U(t, t_0)\,|\psi(t_0)\rangle\,, \qquad U(t, t_0) = e^{-iH(t-t_0)} \tag{114}$$

Propagator kernel as a matrix element.

$$\psi(q, t) = \langle q|\psi(t)\rangle = \int \mathrm{d}q_0\, K(q, t; q_0, t_0)\, \psi(q_0, t_0). \tag{115}$$

Feynman's central insight: the kernel is a sum over all paths, weighted by e^{iS}.

$$K(q, t; q_0, t_0) = \langle q|U(t, t_0)|q_0\rangle = \mathcal{N} \int \mathcal{D}q\, e^{iS} \tag{116}$$

The action and path integral measure.

$$S = \int_{t_0}^{t} \mathrm{d}t\, L(q, \dot{q}), \qquad \int \mathcal{D}q = \lim_{\substack{n\to\infty \\ \Delta t\to 0}} \prod_{m=1}^{n-1} \int \mathrm{d}q(t_m) \tag{117}$$

Normalization factor for a free particle of mass m.

$$\mathcal{N} = \lim_{n\to\infty} \left(\frac{m}{2\pi i \Delta t}\right)^{n/2} \tag{118}$$

In field theory the same idea lifts to functional integrals over field configurations; the vacuum expectation values of time–ordered products are then generated by

$$\langle \Omega|T\phi(x_1)\cdots\phi(x_n)|\Omega\rangle = \frac{\int \mathcal{D}\phi\, \phi(x_1)\cdots\phi(x_n)\, e^{i\int \mathrm{d}^4x \mathcal{L}}}{\int \mathcal{D}\phi\, e^{i\int \mathrm{d}^4x \mathcal{L}}} \tag{119}$$

This provides an alternative derivation of the Feynman rules already obtained via Wick's theorem in Section 5.5 and used in Section 5.6.

6 Gauge Theory and Yang–Mills Fields

The guiding principle of the Standard Model is local gauge symmetry: the dynamics of all fundamental interactions is uniquely dictated by requiring invariance under a continuous local Lie group. This chapter follows Peskin & Schroeder Ch. 15–17.

6.1 General Framework

6.1.1 Lie Algebra Conventions

Hermitian generators T^a ($a = 1, \ldots, N$, where N is the dimension of the gauge group) satisfy the Lie algebra

$$[T^a, T^b] = i\, f^{abc}\, T^c, \tag{120}$$

where f^{abc} are the totally antisymmetric structure constants. They vanish for Abelian groups (e.g. U(1)) and are non–zero for non–Abelian groups (e.g. SU(N)). The fundamental–representation normalization (Peskin convention) is

$$\mathrm{tr}\left[T^a T^b\right] = \frac{1}{2}\delta^{ab}. \tag{121}$$

6.1.2 Covariant Derivative and Field Strength

The covariant derivative is constructed so that $D_\mu\psi(x)$ transforms covariantly under a local gauge transformation $\psi(x) \to U(x)\psi(x)$:

$$D_\mu = \partial_\mu - ig\, T^a A^a_\mu(x), \tag{122}$$

where g is the gauge coupling and A^a_μ are the gauge fields.
The field strength tensor is the commutator of covariant derivatives:

$$F_{\mu\nu} = \frac{i}{g}[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu], \tag{123}$$

or in components,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\, f^{abc} A^b_\mu A^c_\nu. \tag{124}$$

The last term is the self–interaction, absent in the Abelian case.

6.1.3 Yang–Mills Lagrangian and Equations of Motion

The pure gauge–field Lagrangian is Lorentz–invariant, gauge–invariant, and renormalizable:

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu}. \tag{125}$$

Varying the action gives the Yang–Mills equation of motion:

$$D_\mu F^{a\mu\nu} = j^{a\nu}, \tag{126}$$

where $j^{a\nu}$ is the conserved matter current associated with the gauge group. The Bianchi identity follows from the definition of $F_{\mu\nu}$:

$$D_\mu F^a_{\nu\rho} + D_\nu F^a_{\rho\mu} + D_\rho F^a_{\mu\nu} = 0, \tag{127}$$

the non–Abelian generalization of the homogeneous Maxwell equations.

6.2 U(1): Quantum Electrodynamics

QED is the Abelian special case. The gauge group is U(1), with $f^{abc} = 0$ and a single generator (the electric charge Q).

- Gauge field: photon $A_\mu(x)$.
- Covariant derivative (for electrons, $Q = -1$): $D_\mu = \partial_\mu + ieA_\mu$.
- Field strength: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (no self–interaction).
- Full Lagrangian:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi. \tag{128}$$

The photon carries no charge, the field equations are linear, and vacuum polarization screens charge (the coupling increases at high energy).

6.3 SU(2)$_L$: Weak Isospin

The weak interaction acts only on left–handed fermions, which form SU(2)$_L$ doublets.

- Structure constants: $f^{abc} = \varepsilon^{abc}$ (Levi–Civita symbol).
- Generators: $T^a = \sigma^a/2$, $a = 1, 2, 3$.
- Gauge fields: $W^1_\mu, W^2_\mu, W^3_\mu$.
- Covariant derivative:

$$D_\mu = \partial_\mu - ig\,\frac{\sigma^a}{2} W^a_\mu. \tag{129}$$

- Field strength:

$$W^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu + g\,\varepsilon^{abc} W^b_\mu W^c_\nu. \tag{130}$$

The Pauli matrices (already used in the Dirac representation, Ch. 3) are

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{131}$$

After spontaneous symmetry breaking via the Higgs mechanism, W^1_μ, W^2_μ mix into the charged $W^\pm$ bosons, and W^3_μ mixes with the hypercharge field into the Z boson and the photon.

6.4 $\mathrm{SU}(3)_c$: Quantum Chromodynamics

The strong interaction is described by SU(3) color gauge theory.

- Generators: $T^a = \lambda^a/2$, $a = 1, \ldots, 8$, where λ^a are the Gell–Mann matrices.
- Gauge fields: eight gluons $G^a_\mu(x)$.
- Covariant derivative:
$$D_\mu = \partial_\mu - i g_s \frac{\lambda^a}{2} G^a_\mu. \tag{132}$$
- Field strength:
$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s\, f^{abc} G^b_\mu G^c_\nu. \tag{133}$$
- Full QCD Lagrangian:
$$\mathcal{L}_{\mathrm{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + \sum_q \bar{q}_i (i\gamma^\mu D_\mu - m_q) q_i, \tag{134}$$
where i is the color index and q runs over quark flavors.

The eight Gell–Mann matrices are the 3×3 Hermitian generators of SU(3):

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{135}$$

$$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \tag{136}$$

$$\lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{137}$$

Gluons carry color charge, so they interact among themselves (three– and four–gluon vertices). QCD exhibits asymptotic freedom (the coupling weakens at high energy, so quarks and gluons are nearly free) and color confinement (at low energy the coupling is strong and colored particles cannot exist in isolation).

6.5 $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$: Electroweak Unification

The electroweak theory is based on the product group $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, with three weak–isospin generators T^a and one weak–hypercharge generator Y.

- Gauge fields: W^a_μ $(a = 1, 2, 3)$ and B_μ.
- Covariant derivative:
$$D_\mu = \partial_\mu - i g \frac{\sigma^a}{2} W^a_\mu - i \frac{g'}{2} Y\, B_\mu. \tag{138}$$

After Higgs symmetry breaking, the physical gauge fields are linear combinations. The charged bosons are

$$W_\mu^\pm = \frac{1}{\sqrt{2}} \left(W_\mu^1 \mp i W_\mu^2 \right), \tag{139}$$

and the neutral gauge bosons mix through the weak mixing angle θ_W, with $\tan\theta_W = g'/g$:

$$\begin{cases} Z_\mu = \cos\theta_W \, W_\mu^3 - \sin\theta_W \, B_\mu, \\ A_\mu = \sin\theta_W \, W_\mu^3 + \cos\theta_W \, B_\mu. \end{cases} \tag{140}$$

Here A_μ is the massless photon and Z_μ is the massive neutral weak boson.

6.6 Summary

Gauge group	Interaction	# generators	Self–interaction	Key feature
U(1)	Electromagnetic	1	No	Linear, long–range, charge sc
$SU(2)_L$	Weak	3	Yes	Chiral, spontaneously broken, sh
$SU(3)_c$	Strong (QCD)	8	Yes	Asymptotic freedom, color con
$SU(2)_L \times U(1)_Y$	Electroweak	$3+1$	Yes	Higgs mixing, $W^\pm, Z, \gamma$